A First Course in Abstract Algebra

Section 15: Factor-Group Computations and Simple Groups

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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15 Factor-Group Computations and Simple Groups

Exercise 15.1: Removing an entire direct factor

The projection $(a, b) \mapsto a$ from $\mathbb{Z}_2 \times \mathbb{Z}_4$ onto $\mathbb{Z}_2$ has kernel $\{0\} \times \mathbb{Z}_4 = \langle (0, 1) \rangle$. The first isomorphism theorem therefore identifies the quotient with $\mathbb{Z}_2$.

Exercise 15.2: Reducing the second coordinate modulo two

The map $(a, b) \mapsto (a, b \bmod 2)$ is a surjective homomorphism onto $\mathbb{Z}_2 \times \mathbb{Z}_2$. Its kernel is $\{(0, 0), (0, 2)\} = \langle (0, 2) \rangle$. Hence this is the quotient's classification.

Exercise 15.3: A diagonal relation changes the quotient

Define $\phi : \mathbb{Z}_2 \times \mathbb{Z}_4 \to \mathbb{Z}_4$ by $\phi(a, b) = 2a + b \pmod 4$. It is well defined because changing a by 2 changes the result by 4, and it is onto since $\phi(0, 1) = 1$. Its kernel is exactly $(0, 0), (1, 2)$. Thus the quotient by $\langle (1, 2) \rangle$ is $\mathbb{Z}_4$, rather than $\mathbb{Z}_2 \times \mathbb{Z}_2$.

Exercise 15.4: A cyclic quotient of order eight

The surjective map $(a, b) \mapsto b - 2a \pmod 8$ from $\mathbb{Z}_4 \times \mathbb{Z}_8$ to $\mathbb{Z}_8$ is well defined. Its kernel consists exactly of $(a, 2a)$, the subgroup generated by $(1, 2)$. Therefore the quotient is $\mathbb{Z}_8$. Its order is also $32/4 = 8$, checking the computation.

Exercise 15.5: Eliminating one coordinate in a triple product

Map $\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_8$ onto $\mathbb{Z}_4 \times \mathbb{Z}_8$ by

$$\phi(a, b, c) = (b - 2a \pmod 4,\ c - 4a \pmod 8).$$

Changing the representative for a by 4 adds multiples of 4 and 8 to the outputs, so the formula is well defined. It is onto using $a = 0$. Its kernel is $(a, 2a, 4a)$, precisely $\langle (1, 2, 4) \rangle$. Thus the quotient is $\mathbb{Z}_4 \times \mathbb{Z}_8$, of order $128/4 = 32$.

Exercise 15.6: An infinite coordinate projection

The map $\mathbb{Z}^2 \to \mathbb{Z}$, $(a, b) \mapsto a$, is onto with kernel $\langle (0, 1) \rangle$. Hence the quotient is the infinite cyclic group $\mathbb{Z}$.

Exercise 15.7: A primitive diagonal in the integer lattice

Use $\phi(a, b) = b - 2a$. This homomorphism onto $\mathbb{Z}$ has kernel the pairs $(a, 2a)$, exactly $\langle (1, 2) \rangle$. Thus the quotient is $\mathbb{Z}$, with no finite torsion factor.

Exercise 15.8: A primitive diagonal in three dimensions

The map $\phi : \mathbb{Z}^3 \to \mathbb{Z}^2$ given by $(a, b, c) \mapsto (b - a, c - a)$ is onto and has kernel the triples (a, a, a). Therefore the quotient by $\langle (1, 1, 1) \rangle$ is $\mathbb{Z} \times \mathbb{Z}$.

Exercise 15.9: A finite relation on an infinite coordinate

Reducing the first integer coordinate modulo 3 gives the quotient $\mathbb{Z}_3 \times \mathbb{Z} \times \mathbb{Z}_4$. Since 3 and 4 are coprime, $\mathbb{Z}_3 \times \mathbb{Z}_4 \cong \mathbb{Z}_{12}$ by the Chinese remainder theorem. In invariant-factor form the answer is $\mathbb{Z} \times \mathbb{Z}_{12}$. The middle free coordinate remains infinite.

Exercise 15.10: A fourth-multiple relation

The relation only reduces the second integer coordinate modulo 4. The map $(a, b, c) \mapsto (a, b \bmod 4, c)$ is onto $\mathbb{Z} \times \mathbb{Z}_4 \times \mathbb{Z}_8$ with kernel $\langle (0, 4, 0) \rangle$. Thus that direct product is the classification, and $4 \mid 8$ puts the finite factors in invariant-factor order.

Exercise 15.11: A nonprimitive diagonal

Map $\mathbb{Z}^2$ onto $\mathbb{Z} \times \mathbb{Z}_2$ by $\phi(a, b) = (a - b, b \bmod 2)$. Given $(u, \bar{v})$, choose an integer representative v and use $(u + v, v)$, proving surjectivity. The kernel requires $a = b$ and b even, hence is $\langle (2, 2) \rangle$. Therefore the quotient is $\mathbb{Z} \times \mathbb{Z}_2$. The factor of 2 in the diagonal produces the torsion.

Exercise 15.12: A triple diagonal with torsion

The map $(a, b, c) \mapsto (b - a, c - a, a \bmod 3)$ is onto $\mathbb{Z}^2 \times \mathbb{Z}_3$. Its kernel consists of $a = b = c$ with $3 \mid a$, exactly $\langle (3, 3, 3) \rangle$. Thus the quotient is $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}_3$.

Exercise 15.13: Center and commutators of the square group

Write $D_4 = \langle r, s \rangle$ with $r^4 = s^2 = e$ and $srs = r^{-1}$. A rotation r^k commutes with s exactly when $r^k = r^{-k}$, meaning k even; no reflection commutes with r, since that would give $r = r^{-1}$. Thus $Z(D_4) = \{e, r^2\}$. The commutator $rsr^{-1}s^{-1} = r^2$ lies in C. On the other hand the quotient by $\langle r^2 \rangle$ has $r^{-1} = r$, so its defining generators commute and it is abelian. The commutator subgroup is therefore contained in $\langle r^2 \rangle$ as well, proving $C = \{e, r^2\}$.

Exercise 15.14: A direct product with a symmetric group

For any two groups, the center of their product is the product of their centers: commuting is checked in each coordinate. Likewise $[G \times H, G \times H] = [G, G] \times [H, H]$, since commutators are coordinatewise and commutators in either individual coordinate are obtained by setting the other coordinate to the identity. As $\mathbb{Z}_3$ is abelian, $Z(S_3) = \{e\}$, and $[S_3, S_3] = A_3$, the answers are

$$Z(\mathbb{Z}_3 \times S_3) = \mathbb{Z}_3 \times \{e\}, \qquad C = \{0\} \times A_3 \cong \mathbb{Z}_3.$$

For the last standard fact, S_3/A_3 is abelian, while the commutator of two different transpositions is a nonidentity three-cycle, generating A_3.

Exercise 15.15: Two nonabelian direct factors

Apply the coordinatewise center and commutator rules proved in Exercise 15.14. Since $Z(S_3) = \{e\}$, $Z(D_4) = \{e, r^2\}$, $[S_3, S_3] = A_3$, and $[D_4, D_4] = \langle r^2 \rangle$,

$$Z(S_3 \times D_4) = \{e\} \times \langle r^2 \rangle \cong \mathbb{Z}_2, \qquad C = A_3 \times \langle r^2 \rangle \cong \mathbb{Z}_6.$$

The last isomorphism uses the coprime orders 3 and 2.

Exercise 15.16: Every small subgroup of a sixteen-element group

Let $G = \mathbb{Z}_4 \times \mathbb{Z}_4$. Its complete list of subgroups of order at most 4, with quotients, is

subgroup	order	G/H
$\{(0,0)\}$	1	$\mathbb{Z}_4 \times \mathbb{Z}_4$
$\langle (2,0) \rangle$	2	$\mathbb{Z}_2 \times \mathbb{Z}_4$
$\langle (0,2) \rangle$	2	$\mathbb{Z}_2 \times \mathbb{Z}_4$
$\langle (2,2) \rangle$	2	$\mathbb{Z}_2 \times \mathbb{Z}_4$
$\langle (1,0) \rangle,\ \langle (0,1) \rangle$	4	$\mathbb{Z}_4$ (each)
$\langle (1,1) \rangle,\ \langle (1,2) \rangle$	4	$\mathbb{Z}_4$ (each)
$\langle (1,3) \rangle,\ \langle (2,1) \rangle$	4	$\mathbb{Z}_4$ (each)
$\{(0,0), (2,0), (0,2), (2,2)\}$	4	$\mathbb{Z}_2 \times \mathbb{Z}_2$

Here each bracket denotes all multiples of its displayed generator, so the actual subgroup elements are explicit from that rule. To verify completeness, exactly three nonzero elements have order 2. The remaining twelve nonzero elements have order 4, with two generators per cyclic subgroup, giving exactly six cyclic subgroups as listed. A noncyclic order-four subgroup must consist of the identity and all three order-two elements, so there is exactly one more. For an order-two subgroup, the quotient has order 8, exponent at most 4, and an element of order 4: use a coordinate not killed by that subgroup, or $(1,0) + H$ for $H = \langle (2,2) \rangle$. Its classification is therefore $\mathbb{Z}_2 \times \mathbb{Z}_4$. Each cyclic order-four generator has an odd coordinate; elementary invertible changes of coordinates send it to $(1,0)$, so its quotient is $\mathbb{Z}_4$. The remaining subgroup is $2G$, and reducing both coordinates modulo 2 gives its quotient $\mathbb{Z}_2^2$.

Exercise 15.17: The center must contain exactly the commuting elements

Replace "contains all" by "is the set of all." Precisely,

$$Z(G) = \{z \in G : zg = gz \text{ for every } g \in G\}.$$

Saying only that it contains these elements would allow extraneous elements and would not determine a subgroup uniquely.

Exercise 15.18: The commutator subgroup requires generation

The commutator subgroup is the subgroup *generated* by all commutators, not merely their set. It is

$$C = \langle a^{-1}b^{-1}ab : a, b \in G \rangle.$$

This convention gives the same subgroup as $aba^{-1}b^{-1}$, by replacing a, b with their inverses. Individual commutators need not themselves be closed under multiplication, so generation is essential.

Exercise 15.19: Ten statements about factor and simple groups

(a) True. If a generates G, then aN generates every quotient G/N.

(b) False. $\mathbb{Z}_2^2$ is not cyclic, but quotienting by one coordinate gives $\mathbb{Z}_2$.

(c) False. $1/2 + \mathbb{Z}$ has order 2.

(d) True. $1/n + \mathbb{Z}$ has order n, including the identity when $n = 1$.

(e) False. An element killed by 4 must have representative $k/4$. Modulo integers there are only four such classes, and exactly two have order 4: $1/4 + \mathbb{Z}, 3/4 + \mathbb{Z}$.

(f) True. If every commutator is identity, $aba^{-1}b^{-1} = e$ gives $ab = ba$.

(g) False. The inclusion is $C \subseteq H$, because all commutators become identity in an abelian quotient.

(h) False. A simple cyclic group of prime order is abelian and has trivial commutator subgroup.

(i) True. C is normal; it is nontrivial in a nonabelian group, so simplicity forces $C = G$.

(j) False. A_5 is simple and has order 60, which is not prime.

Exercise 15.20: Functions modulo constants

Let $H = \{f : \mathbb{R} \to \mathbb{R} : f(0) = 0\}$ under addition. Define $T : F \to H$ by $T(f)(x) = f(x) - f(0)$. This is an additive homomorphism and is onto because it fixes every function in H. Its kernel is exactly the constant functions K. Thus $F/K \cong H$. Choosing a basepoint turns each coset into its unique representative vanishing there.

Exercise 15.21: Nonvanishing functions modulo constant factors

Let $H^* = \{f : \mathbb{R} \to \mathbb{R} \setminus \{0\} : f(0) = 1\}$ under multiplication. The map $T(f)(x) = f(x)/f(0)$ is well defined and multiplicative, and fixes every function in H^*, so is onto. Its kernel consists exactly of the nonzero constant functions. Therefore $F^*/K^* \cong H^*$.

Exercise 15.22: No order-two additive coset modulo continuous functions

No. If $f + K$ were killed by 2, then $2f$ would be continuous. Multiplying it by the constant $1/2$ would make f continuous as well, so $f \in K$ and its coset is identity, not an element of order 2. The same argument shows the additive quotient has no nonidentity element of any finite order.

Exercise 15.23: An order-two multiplicative coset

Yes. Define $f(x) = 1$ for $x \geq 0$ and $f(x) = -1$ for $x < 0$. It never vanishes but is discontinuous at 0, so $f \notin K^*$. Yet f^2 is the constant function 1 and lies in K^*. Thus fK^* is nonidentity and its square is identity, giving order exactly 2. Unlike scalar multiplication, squaring loses the discontinuous sign information.

Exercise 15.24: A translated full circle

Since $z_0 \in U$, multiplication by z_0 maps U onto U: for any $w \in U$, $w = z_0(z_0^{-1}w)$ and $z_0^{-1}w \in U$. Hence $z_0U = U$, which is certainly a subgroup. Its quotient $U/z_0U = U/U$ is the trivial group.

Exercise 15.25: The circle modulo a half-turn

The homomorphism $U \to U$, $z \mapsto z^2$, is onto because every unit complex number has a unit square root. Its kernel is $\{1, -1\} = \langle -1 \rangle$. Thus $U/\langle -1 \rangle \cong U$ by the first isomorphism theorem.

Exercise 15.26: The circle modulo the nth roots

The map $U \to U$, $z \mapsto z^n$, is a surjective homomorphism. Its kernel is the cyclic subgroup of all nth roots of unity, generated by ζ_n. Consequently $U/\langle \zeta_n \rangle \cong U$. When $n = 1$ this is the identity-map case.

Exercise 15.27: Real numbers modulo integers

The map $\phi : \mathbb{R} \to U$ given by $\phi(t) = \cos(2\pi t) + i \sin(2\pi t)$ is a homomorphism from addition to multiplication, by the angle-addition formulas. It is onto, and $\phi(t) = 1$ exactly when $t \in \mathbb{Z}$. Thus $\mathbb{R}/\mathbb{Z} \cong U$.

Exercise 15.28: Torsion created by quotienting

Take the torsion-free additive group $G = \mathbb{Z}$ and $H = 2\mathbb{Z}$. In G, $na = 0$ with $n > 0$ forces $a = 0$, so no nonzero element has finite order. But $G/H \cong \mathbb{Z}_2$, and all its elements have finite order. Alternatively $\mathbb{Q}/\mathbb{Z}$ gives an infinite quotient with the same property.

Exercise 15.29: Isomorphic kernels with different quotients

Let $G = \mathbb{Z} \times \mathbb{Z}_2$, $H = \mathbb{Z} \times \{0\}$, and $K = 2\mathbb{Z} \times \{0\}$. They are normal because G is abelian, and both are infinite cyclic, hence isomorphic. However $G/H \cong \mathbb{Z}_2$, while $G/K \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. Their different orders show the quotients are not isomorphic.

Exercise 15.30: Centers of simple groups

(a) If a simple group is abelian, every element commutes with every other, so $Z(G) = G$. A nontrivial simple abelian group is cyclic of prime order.

(b) If it is nonabelian, its normal center cannot be G. Simplicity leaves only $Z(G) = \{e\}$.

Exercise 15.31: Commutators of simple groups

(a) In an abelian group, every commutator is e, so $C = \{e\}$.

(b) In a nonabelian group at least one commutator is nonidentity. Thus C is a nontrivial normal subgroup, and simplicity forces $C = G$.

Exercise 15.32: Synopsis of cyclic quotients

If $G = \langle a \rangle$, then every coset is $a^k N = (aN)^k$, so aN generates G/N. This one observation proves the cyclic-quotient theorem.

Exercise 15.33: Synopsis of maximal normal subgroups

Under the quotient map $G \to G/M$, normal subgroups properly between M and G correspond exactly to nontrivial proper normal subgroups of G/M. Thus absence of the former, meaning M is maximal normal, is equivalent to simplicity of G/M.

Exercise 15.34: Index two prevents simplicity

A subgroup H of index 2 is normal: for $g \notin H$, both gH and Hg are the complement of H, while for $g \in H$ both equal H. The subgroup is proper because its index is 2 and nontrivial by hypothesis. It is therefore a nontrivial proper normal subgroup of G, so G is not simple.

Exercise 15.35: The image of a normal subgroup

The image $\phi[N]$ is a subgroup of $\phi[G]$. For $\phi(g) \in \phi[G]$ and $\phi(n) \in \phi[N]$,

$$\phi(g)\phi(n)\phi(g)^{-1} = \phi(gng^{-1}) \in \phi[N],$$

since N is normal in G. Thus the image is normal in $\phi[G]$. It need not be normal in all of G' when ϕ is not onto.

Exercise 15.36: The inverse image of a normal subgroup

Set $H = \phi^{-1}[N']$. It is a subgroup because identity, products and inverses map into the subgroup N'. For $g \in G, h \in H$,

$$\phi(ghg^{-1}) = \phi(g)\phi(h)\phi(g)^{-1} \in N',$$

by normality of N' in G'. Hence $ghg^{-1} \in H$, proving normality in G. No surjectivity assumption is needed.

Exercise 15.37: A cyclic central quotient forces commutativity

Prove the contrapositive. Suppose $G/Z(G)$ is generated by $aZ(G)$. Each $x \in G$ then has form $a^m z$ with $z \in Z(G)$, and similarly $y = a^n w$. Since the central factors commute with all elements,

$$xy = a^{m+n} zw = a^{n+m} wz = yx.$$

Thus G is abelian. Therefore a nonabelian group cannot have a cyclic quotient by its center.

Exercise 15.38: A nonabelian group of order a product of primes

The center is a subgroup, so its order divides pq. If it were the whole group, G would be abelian. If it were nontrivial and proper, its order would be p or q (also when $p = q$). The quotient would then have prime order and hence be cyclic. Exercise 15.37 would make G abelian, again a contradiction. Thus $Z(G) = \{e\}$.

Exercise 15.39: A complete proof of simplicity of the alternating group

We compose permutations from right to left. All letters appearing distinctly in a displayed identity are assumed distinct. Put $c_i = (r, s, i)$ for fixed distinct r, s and $i \notin \{r, s\}$. The source's description of "n special cycles" means these $n - 2$ actual three-cycles; repeated entries $i = r, s$ are not cycles.

(a) A three-cycle is a product of two transpositions, for example $(a, b, c) = (a, c)(a, b)$, so is even and lies in A_n.

(b) Every even permutation is a product of an even number of transpositions. Group them into adjacent pairs. Equal transpositions cancel. A pair sharing one letter is a three-cycle (or its inverse). For a disjoint pair, direct computation gives $(a, b)(c, d) = (a, c, b)(a, c, d)$. Thus every such pair is a product of three-cycles, and these generate A_n.

(c) To show the special three-cycles suffice, compute

$$c_i^2 = (r, i, s), \qquad c_j c_i^2 = (r, i, j),$$
$$c_j^2 c_i = (s, i, j), \qquad c_i^2 c_k c_j^2 c_i = (i, j, k).$$

Their inverses give the opposite orientations. These cover three-cycles containing both r, s, only r, only s, or neither. Thus every three-cycle is a product of the special ones, and (b) proves generation. If $n = 3$, only c_i, c_i^{-1} are needed.

(d) Suppose $N \trianglelefteq A_n$ contains a three-cycle, relabeled as c_i. Then $c_i^2 \in N$. For any other $j \notin \{r, s, i\}$, the double transposition $h = (r, s)(i, j)$ is even, and conjugation gives $hc_i^2 h^{-1} = (r, s, j) = c_j \in N$. Thus N contains all special cycles (and c_i itself). Part (c) gives $N = A_n$. For $n = 3$ the contained three-cycle already generates A_3.

(e) Take a nonidentity $\sigma \in N$ and use its disjoint-cycle decomposition. We check every case in the source. For any $t \in A_n$, normality gives $\sigma^{-1} t\sigma t^{-1} \in N$: the first factor σ^{-1} and the conjugate $t\sigma t^{-1}$ both belong to N.

Case I: N contains a three-cycle as an element.

Part (d) applies immediately. Merely having a three-cycle among the disjoint factors of a larger permutation does not suffice; those cases are treated below.

Case II: there is a cycle of length $\ell > 3$.

Write $\sigma = \mu(a_1, a_2, \ldots, a_\ell)$, with μ disjoint from that cycle. Set $t = (a_1, a_2, a_3)$. Conjugation replaces the entries by their preimages under σ, so

$$\sigma^{-1}t\sigma t^{-1} = (a_\ell, a_1, a_2)(a_1, a_3, a_2) = (a_1, a_3, a_\ell).$$

This is a three-cycle in N, so (d) applies.

Case III: at least two three-cycles occur.

Write $\sigma = \mu(a_4, a_5, a_6)(a_1, a_2, a_3)$, all factors disjoint, and choose $t = (a_1, a_2, a_4)$. Then

$$\sigma^{-1}t\sigma t^{-1} = (a_3, a_1, a_6)(a_1, a_4, a_2) = (a_1, a_4, a_2, a_6, a_3).$$

This five-cycle lies in N, reducing to Case II, hence to Case I.

Case IV: precisely one three-cycle and some two-cycles occur.

Here $\sigma = \mu(a_1, a_2, a_3)$ with μ a disjoint product of transpositions. Since $\mu^2 = e$ and disjoint factors commute, $\sigma^2 = (a_1, a_3, a_2) \in N$, a three-cycle.

Case V: all nontrivial cycles are transpositions.

Since σ is even, there are an even number of them, at least two. Write $\sigma = \mu(a_3, a_4)(a_1, a_2)$ with μ disjoint. Taking $t = (a_1, a_2, a_3)$ gives

$$\sigma^{-1}t\sigma t^{-1} = (a_2, a_1, a_4)(a_1, a_3, a_2) = (a_1, a_3)(a_2, a_4) = \alpha \in N.$$

Because $n \geq 5$, choose a letter j different from these four and let $\beta = (a_1, a_3, j) \in A_n$. Another conjugation calculation gives

$$\beta^{-1}\alpha\beta\alpha = (j, a_1)(a_2, a_4)(a_1, a_3)(a_2, a_4) = (a_1, a_3, j) \in N.$$

This is again a three-cycle. These cases exhaust all possible nonidentity even permutations: a longer cycle gives II; otherwise the number of three-cycles is at least two, exactly one, or zero, giving III, IV, or V. Every nontrivial normal subgroup is therefore A_n, proving simplicity for $n \geq 5$.

Exercise 15.40: The product of a subgroup and a normal subgroup

The set HN contains both subgroups. For $h_1n_1, h_2n_2 \in HN$, normality gives

$$(h_1n_1)(h_2n_2) = h_1h_2(h_2^{-1}n_1h_2)n_2 \in HN.$$

Also $(hn)^{-1} = n^{-1}h^{-1} = h^{-1}(hn^{-1}h^{-1}) \in HN$. Together with the identity this proves HN is a subgroup. Any subgroup containing both H and N contains every product hn, so contains HN. Thus HN is the smallest such subgroup.

Exercise 15.41: A product of two normal subgroups

By Exercise 15.40, NM is a subgroup. If $g \in G$ and $nm \in NM$, then $g(nm)g^{-1} = (gng^{-1})(gmg^{-1})$. The two factors lie in N and M, respectively, by their normality, so the product lies in NM. Therefore NM is normal in G.

Exercise 15.42: Disjoint normal subgroups commute elementwise

For $h \in H, k \in K$, let $c = hkh^{-1}k^{-1}$. It belongs to K, because $hkh^{-1} \in K$ and $k^{-1} \in K$. It also belongs to H, because $kh^{-1}k^{-1} \in H$ and $c = h(kh^{-1}k^{-1})$. The intersection hypothesis gives $c = e$, so $hk = kh$. Thus all pairs of elements from the two subgroups commute.